ELEC2104/9704 Tutorial 6 Solutions (2025)

For any BJT circuit analysis question, the first step is label all BJT voltages and currents and with correct polarities/directions. For NPN transistors, the arrow points *outward* from the emitter, indicating conventional current flow direction. For PNP transistors, the arrow points *inward* to the emitter, with opposite current directions.

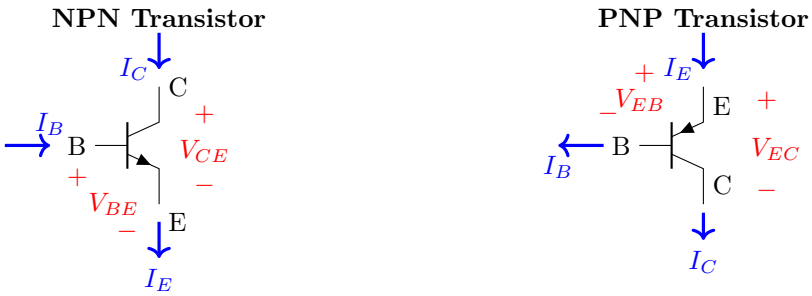


Figure 1: NPN and PNP BJT symbols with proper voltage and current labeling conventions

Technique to establish the correct BJT current directions and voltage polarities
Observe the arrow direction in the BJT circuit symbol. This arrow indicates the emitter E terminal and the direction of current flow for the emitter current I_E. The remaining currents I_C and I_B have the opposite direction. For example, if I_E points outward (NPN), then I_C and I_B point inward. The BJT voltage directions are given by these current directions, as the current arrows point from the positive (+) to the negative (-) of their corresponding voltages.

Essential BJT Equations for Circuit Analysis

Fundamental BJT equations you'll use repeatedly in this course:

1. **Exponential I-V Relationship:**

$$I_C = I_S \left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \approx I_S \exp\left(\frac{V_{BE}}{V_T}\right) \quad \text{when } V_{BE} \gg V_T$$

   This describes the non-linear, exponential relationship between collector current and base-emitter voltage. Small changes in V_{BE} cause large changes in I_C.

2. **Current Relationships:**

$$I_E = I_C + I_B \quad \text{and} \quad I_C = \beta I_B \quad \text{where } \beta \text{ is the current gain (typically 100-200)}$$

3. **Small-Signal Parameters (for AC analysis):**

$$g_m = \frac{I_C}{V_T} \quad \text{(transconductance)} \quad \text{and} \quad r_\pi = \frac{\beta}{g_m} \quad \text{(input resistance)}$$

   These linearized parameters are crucial for analysing BJT amplifiers (next tutorial).

1 Question 1

In the circuit below, it is observed that the collector currents of Q_1 and Q_2 are equal if $V_{BE1} - V_{BE2}$ = 20 mV. Determine the ratio of transistor cross-section areas if the other device parameters are identical. Assume $V_T = 26\,\text{mV}$.

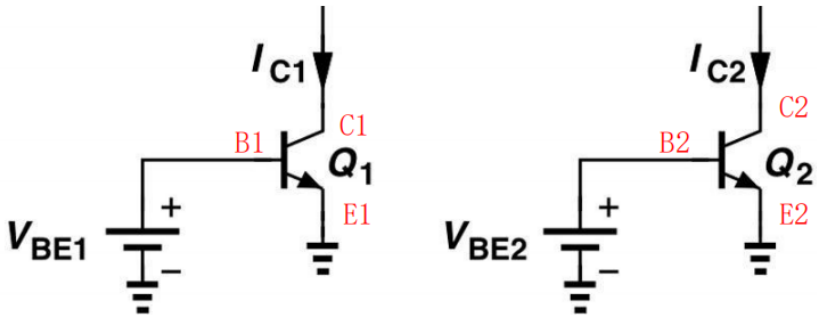


Figure 2: Q1 circuit diagram

Solution:

For a BJT, (in this course) the correlation between the voltage drop between base and emitter (V_{BE}) and the collector current (I_C) is modelled by the exponential relation given by:

$$I_C = I_S \left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \approx I_S \exp\left(\frac{V_{BE}}{V_T}\right) \tag{1}$$

The following formula for BJT saturation current was presented in the lecture:

$$I_S = \frac{AqD_n n_i^2}{N_A W_B} \tag{2}$$

Taking (1) into (2), we find that I_C is proportional to the transistor cross-section area A, so $I_C \propto A$.

$$I_C = \left(\frac{AqD_n n_i^2}{N_A W_B}\right) \exp\left(\frac{V_{BE}}{V_T}\right). \tag{3}$$

Since $I_{C1} = I_{C2}$, we have:

$$\frac{A_1 qD_n n_i^2}{N_A W_B} \exp\left(\frac{V_{BE1}}{V_T}\right) = \frac{A_2 qD_n n_i^2}{N_A W_B} \exp\left(\frac{V_{BE2}}{V_T}\right). \tag{4}$$

Note the subscripts for A_1, V_{BE1} and A_2, V_{BE2} denoting quantities for transistors $Q1$ and $Q2$, respectively. The question states that all other device parameters are identical.

From (4), we can derive the ratio of the two cross-section areas, note that the identical parameters cancelled out:

$$A_1 \exp\left(\frac{V_{BE1}}{V_T}\right) = A_2 \exp\left(\frac{V_{BE2}}{V_T}\right). \tag{5}$$

$$\frac{A_2}{A_1} = \frac{\exp\left(\frac{V_{BE1}}{V_T}\right)}{\exp\left(\frac{V_{BE2}}{V_T}\right)} = \exp\left(\frac{V_{BE1} - V_{BE2}}{V_T}\right). \tag{6}$$

Substituting $V_{BE1} - V_{BE2} = 20\,\text{mV}$ and $V_T = 26\,\text{mV}$, we get the value of the ratio.

$$\frac{A_2}{A_1} = \exp\left(\frac{0.020}{0.026}\right) \approx 2.158. \tag{7}$$

Or:

$$\frac{A_1}{A_2} = \exp\left(\frac{-0.020}{0.026}\right) \approx 0.463. \tag{8}$$

2 Question 2

Consider the circuit below. Assume $V_T = 26\,\text{mV}$.

(a) If $I_{S1} = 2I_{S2} = 5 \times 10^{-16}\,\text{A}$, determine V_B such that $I_X = 1.2\,\text{mA}$.

(b) What value of R_C places the transistors at the edge of the active mode?

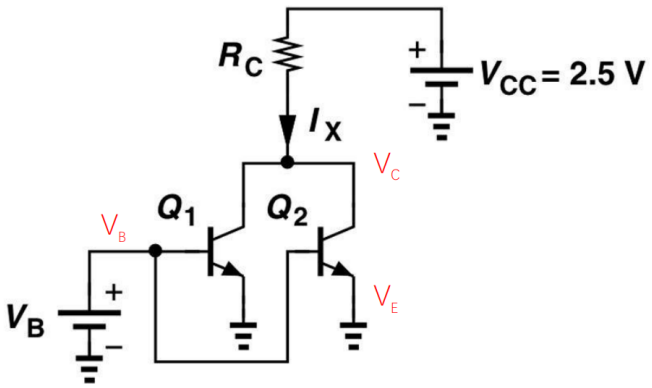


Figure 3: Q2 circuit diagram

2.1 Part A: If $I_{S1} = 2I_{S2} = 5 \times 10^{-16}\,\text{A}$, determine V_B such that $I_X = 1.2\,\text{mA}$

From the circuit diagram, we immediately notice that the two BJTs share a base voltage (V_B) and a collector voltage (V_C). They also have the same emitter voltage (V_E) since both emitters are connected to ground. Therefore, we write $V_{BE1} = V_{BE2} = V_B$.

Next, we write an expression for I_X using KCL on the shared collector node:

$$I_X = I_{C1} + I_{C2} = I_{S1} \exp\left(\frac{V_{BE1}}{V_T}\right) + I_{S2} \exp\left(\frac{V_{BE2}}{V_T}\right) \tag{1}$$

The question gives the following information:

$$\begin{aligned} I_{S1} &= 2I_{S2} = 5 \times 10^{-16}\,\text{A}, \\ I_X &= 1.2\,\text{mA}, \\ V_T &= 26\,\text{mV}. \end{aligned} \tag{2}$$

Using the above, we can solve for V_B by substituting $I_{S2} = \frac{1}{2} I_{S1}$ and $V_{BE1} = V_{BE2} = V_B$:

$$I_X = I_{S1} \exp\left(\frac{V_B}{V_T}\right) + \tfrac{1}{2} I_{S1} \exp\left(\frac{V_B}{V_T}\right) \tag{3}$$

$$1.2 \times 10^{-3}\,\text{A} = \left(1 + \tfrac{1}{2}\right)\left(5 \times 10^{-16}\,\text{A}\right) \exp\left(\frac{V_B}{0.026\,\text{V}}\right) \tag{4}$$

$$1.6 \times 10^{12} = \exp\left(\frac{V_B}{0.026\,\text{V}}\right) \tag{5}$$

$$V_B = \ln\left(1.6 \times 10^{12}\right)(0.026\,\text{V}) = 0.731\,\text{V} \tag{6}$$

Answer: $V_B = 0.731\,\text{V}$

A more detailed, step-by-step walkthrough of this solution is provided in Section 2.3 below.

Important Note: V_{BE} vs V_B

Be careful to distinguish between V_{BE} (base-emitter voltage) and V_B (base voltage with respect to ground).
In this circuit, since both emitters are connected to ground (0V), we have $V_{BE} = V_B - V_E = V_B - 0 = V_B$.
However, if there were a resistor between the emitter and ground, then $V_E \neq 0$ and $V_{BE} = V_B - V_E \neq V_B$. This is a common source of confusion in BJT circuit analysis.

BJT Operating Regions (NPN)

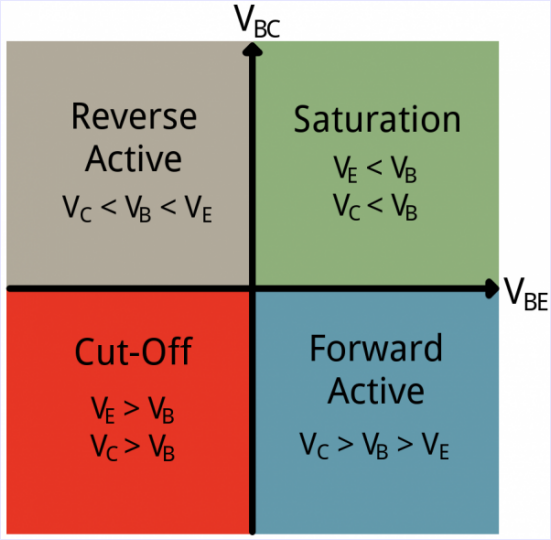


Figure 4: BJT operating regions as a function of V_{BE} and V_{BC} voltages (NPN)

The figure above shows the four operating regions of a BJT transistor based on the base-emitter (V_{BE}) and base-collector (V_{BC}) junction voltages. The table below provides detailed conditions for each region:

Region	**Voltage Relationships (NPN)**
Cut-off	$V_E > V_B$ and $V_C > V_B$ Both junctions reverse-biased
Forward Active (very common)	$V_C > V_B > V_E$ BE forward, BC reverse
Saturation	$V_E < V_B$ and $V_C < V_B$ Both junctions forward-biased
Reverse Active (rarely used)	$V_C < V_B < V_E$ BE reverse, BC forward

For PNP transistors, simply flip the directions of these inequalities.
Key insight for Q2 Part B:

- The boundary between Forward Active and Saturation occurs when $V_{BC} = 0$, which means $V_C = V_B$, that is, the x-axis intercept in Fig 4.
- At this point, the BC junction is about to become forward-biased
- This is the "edge of active mode" condition we need to find R_C

2.2 Part B: What value of R_C places the transistors at the edge of the active mode?

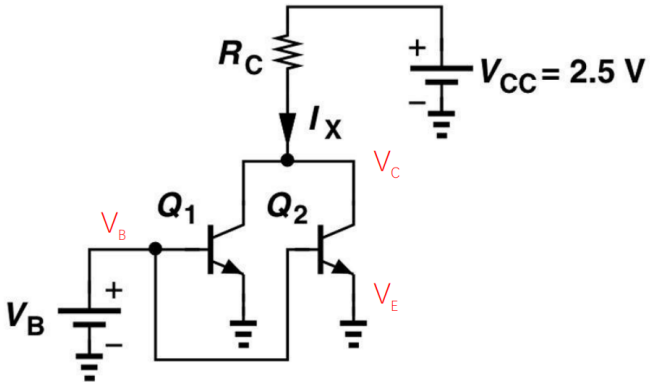


Figure 5: Q2 circuit diagram

To find the value of R_C that places the transistors at the edge of active mode, we need to determine when the transistors are about to enter saturation.

Step 1: Identify the boundary condition
At the boundary between forward-active and saturation regions, the base-collector junction is about to become forward-biased. This occurs when:

$$V_C = V_B \quad \text{(edge of saturation condition)} \tag{7}$$

Step 2: Apply Ohm's law across R_C
The current I_X flows through resistor R_C, creating a voltage drop from V_{CC} to V_C:

$$I_X = \frac{V_{CC} - V_C}{R_C} \tag{8}$$

Step 3: Substitute the boundary condition
Since we want $V_C = V_B$ at the edge of active mode, we substitute:

$$I_X = \frac{V_{CC} - V_B}{R_C} \tag{9}$$

Step 4: Solve for R_C
Rearranging (9) to isolate R_C:

$$R_C = \frac{V_{CC} - V_B}{I_X} \tag{10}$$

Step 5: Substitute known values
From Part A, we found $V_B = 0.731\,\text{V}$. Using the given values:

- $V_{CC} = 2.5\,\text{V}$ (from circuit diagram)
- $V_B = 0.731\,\text{V}$ (calculated in Part A)
- $I_X = 1.2\,\text{mA}$ (given)

$$R_C = \frac{2.5\,\text{V} - 0.731\,\text{V}}{1.2 \times 10^{-3}\,\text{A}} = \frac{1.769\,\text{V}}{1.2 \times 10^{-3}\,\text{A}} = 1474.17\,\Omega \tag{11}$$

Answer: $R_C = 1.47\,\text{k}\Omega$ places both transistors at the edge of active mode ($V_C = V_B$).

2.3 Part A: Detailed Walkthrough

Here we revisit Q2.a using systematic problem-solving techniques.

Variable Inventory Create an inventory of known vs unknown variables to prevent getting lost in complex problems.
Known variables:

- $I_{S1} = 5 \times 10^{-16}\,\text{A}$ (given)
- $I_{S2} = \frac{1}{2} I_{S1} = 2.5 \times 10^{-16}\,\text{A}$ (from given relationship)
- $I_X = 1.2\,\text{mA}$ (given)
- $V_{CC} = 2.5\,\text{V}$ (from circuit diagram)
- $V_T = 26\,\text{mV}$ (thermal voltage at room temperature)
- $V_{E1} = V_{E2} = 0\,\text{V}$ (emitters connected to ground)

Target variable:

- $V_B = V_{BE1} = V_{BE2}$ **(what we're solving for)**

Circuit Analysis Strategy Now let's solve systematically using the *"build-up"* approach:

1. Start by listing the question's given quantities and noting the target variable/s to solve for
2. Identify circuit relationships (KCL, KVL, component laws) to build a *mathematical model* of the circuit
3. Count the number of unknown variables in these relationships and mark the variables that are known.
4. Look for ways to substitute and reduce the number of unknowns - are there circuit relationships you haven't described mathematically yet?
5. Substitute numerical values only at the end when you have the target variable/s in terms of known values.

Key insight: If you get stuck, you may be missing a circuit feature from the math, or a given value from the question. Ask yourself: "What circuit relationships haven't I captured in equations yet?"

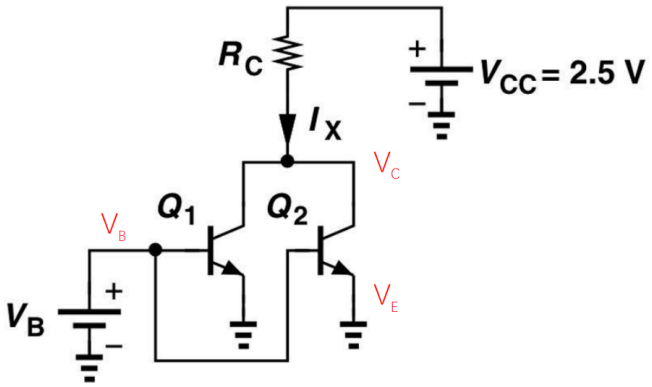


Figure 6: Q2 circuit diagram

We start with $I_X = 1.2\,\text{mA}$ (given) and ask: *"How does this current relate to other circuit elements?"*
Circuit observation: From Fig 6, current I_X flows into *both* BJT collector terminals at their shared connection.
Mathematical translation: This is a classic KCL application! At the shared collector node:

$$\boxed{\text{Current in} = \text{Current out}} \Rightarrow \boxed{I_X = I_{C1} + I_{C2}}$$

Note: Don't assume $I_X = I_{C1} = I_{C2}$. They are different currents related by KCL.
Using KCL at the shared collector node (blue = known, red = unknown):

$$I_X = I_{C1} + I_{C2} \tag{12}$$

Note that we are trying to find either V_B, V_{BE1} or V_{BE2} for this question (they are all equal). Currently, I_{C1} and I_{C2} are unknown, but for BJTs, the collector currents I_{C1} and I_{C2} can easily be related to V_{BE1} and V_{BE2} using the exponential I_C relation:

$$I_C = I_S \left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right] \tag{13}$$

Substituting the above (13) into our expression for I_X in (12) gives us a more direct relation between known variables and the variables we are solving for:

$$I_X = I_{S1}\left[\exp\left(\frac{V_{BE1}}{V_T}\right) - 1\right] + I_{S2}\left[\exp\left(\frac{V_{BE2}}{V_T}\right) - 1\right] \tag{14}$$

Expect to make this substitution for many (but not all) BJT circuits, as making this substitution describes the non-linear behaviour of the BJT in the circuit, capturing this behaviour into the math.

Counting Unknowns: Looking at the color-coding in (14), we initially have two red unknowns: V_{BE1} and V_{BE2}. However, from our circuit analysis, we know that $V_{BE1} = V_{BE2} = V_B$ (both emitters grounded). This reduces our problem to just one unknown, which means we can solve it!

$$I_X = I_{S1}\left[\exp\left(\frac{V_B}{V_T}\right) - 1\right] + \tfrac{1}{2} I_{S1}\left[\exp\left(\frac{V_B}{V_T}\right) - 1\right] \tag{15}$$

In the above, we used $V_B = V_{BE1} = V_{BE2}$ and $I_{S2} = \frac{1}{2} I_{S1}$. It is helpful to simplify the expression now and leave substituting values until the end:

$$\begin{aligned}
I_X &= I_{S1}\left(1 + \frac{1}{2}\right)\left[\exp\left(\frac{V_B}{V_T}\right) - 1\right] \\
\exp\left(\frac{V_B}{V_T}\right) - 1 &= \frac{I_X}{I_{S1}\left(1 + \frac{1}{2}\right)} \\
\exp\left(\frac{V_B}{V_T}\right) &= \frac{I_X}{I_{S1}\left(1 + \frac{1}{2}\right)} + 1 \\
\frac{V_B}{V_T} &= \ln\left(\frac{I_X}{I_{S1}\left(1 + \frac{1}{2}\right)} + 1\right) \\
V_B &= V_T \ln\left(\frac{I_X}{I_{S1}\left(1 + \frac{1}{2}\right)} + 1\right)
\end{aligned} \tag{16}$$

Substituting the known values from the question gives:

$$\begin{aligned}
V_B &= (0.026\ \text{V}) \ln\left(\frac{1.2\ \text{mA}}{(5 \times 10^{-16}\ \text{A})\left(1 + \frac{1}{2}\right)} + 1\right) \\
V_B &= 0.731\ \text{V}
\end{aligned} \tag{17}$$

Engineering Approximation Tip

About the "-1" term in the BJT equation:
Notice we used $I_C = I_S[\exp(V_{BE}/V_T) - 1]$ throughout this section. However, when the transistor is "on" (forward-biased with $V_{BE} \approx 0.7\text{V}$), the exponential term becomes very large:

$$\exp\left(\frac{0.7\text{V}}{0.026\text{V}}\right) = \exp(26.9) \approx 4.9 \times 10^{11}$$

In this case, subtracting 1 from such a large number is negligible:

$$4.9 \times 10^{11} - 1 \approx 4.9 \times 10^{11}$$

Practical rule: When $V_{BE} \gg V_T$ (transistor clearly "on"), we can use the simplified form:

$$I_C \approx I_S \exp\left(\frac{V_{BE}}{V_T}\right)$$

Try it yourself: Re-calculate this problem using the simplified equation without the "-1" term. You should get the same answer within rounding precision! This approximation is commonly used in practical BJT circuit analysis.

3 Question 3

Consider the circuit below, assuming $\beta = 100$, $V_T = 26\,\text{mV}$, and $I_S = 7 \times 10^{-16}\,\text{A}$. If $R_1 = 10\,\text{k}\Omega$, determine V_S such that $I_C = 1\,\text{mA}$.

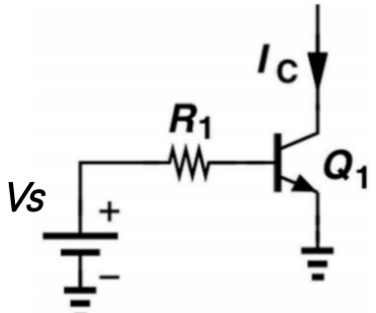


Figure 7: Q3 circuit diagram

Solution:

Key to solving this question is relating the source voltage V_S to the transistor using Ohm's Law on R_1 (see (5)) and finding both I_B and V_B through the standard BJT formulae, given $I_C = 1\,\text{mA}$.

From the BJT current relationships, the base current I_B can be written using the collector I_C and the current gain term β:

$$I_B = \frac{I_C}{\beta} = \frac{1 \times 10^{-3}\,\text{A}}{100} = 1 \times 10^{-5}\,\text{A}. \tag{1}$$

For a BJT, the correlation between the voltage drop between base and emitter (V_{BE}) and the collector current (I_C) is:

$$I_C = I_S \left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right]. \tag{2}$$

Rearranging (2) (and assuming $V_{BE} \gg V_T$ to omit the -1 term), we can get the expression of V_{BE}:

$$V_{BE} = V_T \ln\left(\frac{I_C}{I_S}\right). \tag{3}$$

Substituting the values of I_C, I_S and V_T, the value of V_{BE} is determined:

$$V_{BE} = (0.026\,\text{V}) \cdot \ln\left(\frac{1 \times 10^{-3}\,\text{A}}{7 \times 10^{-16}\,\text{A}}\right) = 0.728\,\text{V}. \tag{4}$$

Next, applying Ohm's Law to the R_1 base resistor, we get:

$$\frac{V_S - V_B}{R_1} = I_B \tag{5}$$

As $V_E = 0\,\text{V}$ (emitter connected to ground), $V_{BE} = V_B - V_E = V_B$.
Rearranging (5) and substituting $V_B = V_{BE}$ and numerical values, we can determine V_S:

$$\begin{aligned} V_S &= V_{BE} + I_B \cdot R_1 \\ &= 0.728\,\text{V} + (1 \times 10^{-5}\,\text{A})(1 \times 10^4\,\Omega) \\ &= 0.828\,\text{V}. \end{aligned} \tag{6}$$

4 Question 4

What is the transconductance of an npn transistor operating at 350 K and a collector current of

(a) 10 μA,

(b) 100 μA,

(c) 1 mA,

(d) 10 mA?

(e) Repeat for a pnp transistor.

Solution:

In a BJT, the transconductance g_m is used to describe the sensitivity of the collector current I_C to changes in the base-emitter voltage V_{BE}. It is defined as:

$$g_m = \frac{I_C}{V_T} \tag{1}$$

The thermal voltage at $T = 350\,\text{K}$ is:

$$V_T = \frac{kT}{q} = \frac{(1.38 \times 10^{-23}\,\text{J/K})(350\,\text{K})}{1.6 \times 10^{-19}\,\text{C}} = 0.0302\,\text{V} \tag{2}$$

4.1 Part A

Taking $I_C = 10\,\mu\text{A}$ and substituting into (1):

$$g_m = \frac{10^{-5}}{0.0302} = 0.33\,\text{mS}. \tag{3}$$

4.2 Part B

For $I_C = 100\,\mu\text{A}$,

$$g_m = \frac{10^{-4}}{0.0302} = 3.31\,\text{mS}. \tag{4}$$

4.3 Part C

For $I_C = 1\,\text{mA}$,

$$g_m = \frac{10^{-3}}{0.0302} = 33.1\,\text{mS}. \tag{5}$$

4.4 Part D

For $I_C = 10\,\text{mA}$,

$$g_m = \frac{10^{-2}}{0.0302} = 331\,\text{mS}. \tag{6}$$

4.5 Part E

The values of g_m are identical for both npn and pnp transistors, since (1) depends only on the collector current I_C and the thermal voltage V_T, and not on transistor polarity.